10 Motor and Drive Control

By Dave Polka

Topic Highlights

DC Motors and Their Principles of Operation
DC Motor Types
AC Motors and Their Principles of Operation
AC Motor Types
Choosing the Right Motor
Variable Speed Drives (Electronic DC)
Variable Speed Drives (Electronic AC)
Automation and the Use of VFDs

10.1 Introduction

In order to truly understand modern day automation, it is vital that a study of motor and electronic drive principles be undertaken. The drive is the device that controls the motor. The two interact to provide the torque, speed, and horsepower (HP) necessary to operate the application.

10.2 DC Motors and Their Principles of Operation

There are two basic circuits in any direct current (DC) motor: the *armature* (device that rotates) and the *field* (stationary part with windings). The two components magnetically interact with one another to produce rotation of the armature. Both the armature and the field are two separate circuits and are physically next to each other, in order to promote magnetic interaction.

The armature (I_A) has an integral part, called a "commutator" (see Figure 10-1). The commutator acts as an electrical switch, always changing polarity of the magnetic flux to ensure there is a "repelling" force taking place. The armature rotates as a result of the "repelling" motion created by the magnetic flux of the armature, in opposition to the magnetic flux created by the field winding (I_F).

The physical connection of voltage to the armature is done through "brushes." Brushes are made of a carbon material that is in constant contact with the armature's commutator plates. The brushes are typically spring loaded to provide constant pressure of the brush to the commutator plates.

10.2.1 Control of Speed

The speed of a DC motor is a direct result of armature voltage applied. The field receives voltage from a separate power supply, sometimes referred to as a "field exciter." This exciter provides power to the field which, in turn, generates current and magnetic flux. In a normal condition, the field is kept at

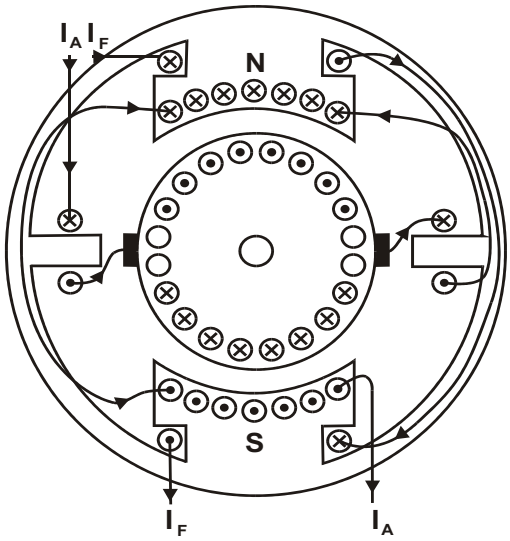


Figure 10-1: Armature and Field Connections (Courtesy of ABB Inc.)

maximum strength, allowing the field winding to develop maximum current and flux (known as the "armature range"). The only way to control the speed is through change in armature voltage.

10.2.2 Control of Torque

Under certain conditions, motor torque remains constant, when operating below base speed. However, when operating in the field weakening range, torque drops off inversely as $1/Speed^2$. If field flux is held constant, as well as the design constant of the motor, then torque is proportional to the armature current. The more load the motor sees, the more current that is consumed by the armature.

10.2.3 Enclosure Types and Cooling Methods

In most cases, to allow the motor to develop full torque at less than 50% speed, an additional blower is required for motor cooling. The enclosures most commonly found in standard industrial applications are:

- DPFG (Drip-proof - Fully Guarded) — This type of enclosure is self-ventilated and has no external means of cooling. Most DPFG designs can generate 100% rated torque down to 50% of base speed (see Figure 10-2).
- TENV (Totally Enclosed Non-Ventilated) — This type of enclosure has no external cooling, but uses an internal fan to circulate the air within the motor. This type of motor is capable of delivering 100% torque down to 10 or 5% of base speed.
- TEFC (Totally Enclosed Fan Cooled) — This type of enclosure has an externally mounted fan on the commutator end shaft. Air flow is a direct result of the speed of the motor, which also means that this type of enclosure is not suitable for low speed applications.

10.2.4 Protection and Ratings

- ***Ambient Temperature*** - Typical recommendations are for the motor ambient conditions not to exceed 40^oC (104^oF). Motors continuously used in higher temperatures will need a lower temperature rise class of insulation. Major insulation temperature classes refer to mechanical and dielectric strength, and are: A (lowest grade), B, F, and H (highest grade).

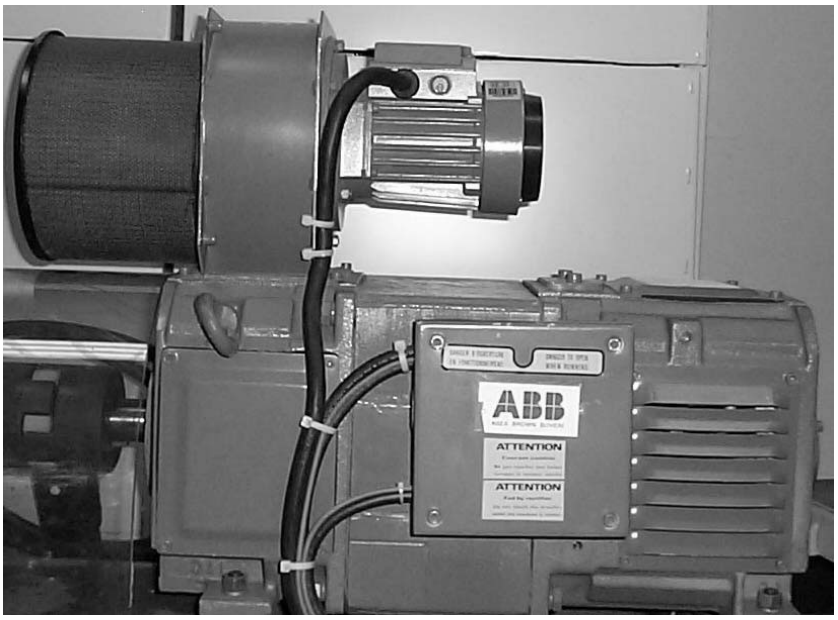


Figure 10-2: DPFG Motor (Courtesy of ABB Inc.)

- ***Over-Temperature Conditions*** - Placing the motor into overload conditions is one cause of over-temperature. High temperature inside the motor causes expansion stress in the wire insulation, resulting in cracks which, in turn, can cause contamination and eventual wire failure.
- ***Nameplate Ratings*** - Typical DC motor nameplate ratings are: Frame, HP, Amps/Field & Armature, Base/Max Speed. Additional ratings include: enclosure type, thermostat type, ambient rating, catalog and serial number, and tachometer, duty ratings.

10.3 DC Motor Types

10.3.1 Series Wound

A series wound DC motor has the armature and field windings connected in a series circuit. Starting torque developed can be as high as 500% of the full load rating. The high starting torque is a result of the field winding being operated below the saturation point. An increase in load will cause a corresponding increase in both armature and field winding current, which means both armature and field winding flux increase together. Torque in a DC motor increases as the square of the current value. Compared to a shunt wound motor, a series wound motor will generate a larger torque increase for a given increase in current.

10.3.2 Parallel (Shunt) Wound

Shunt wound DC motors have the armature and field windings connected in parallel (see Figure 10-3).

This type of motor requires two power supplies—one for the armature and one for the field winding. The starting torque developed can be 250-300% of the full load torque rating, for a short period of time. Speed regulation (speed fluctuation due to load) is acceptable in many cases, between 5-10% of maximum speed, when operated from a DC drive.

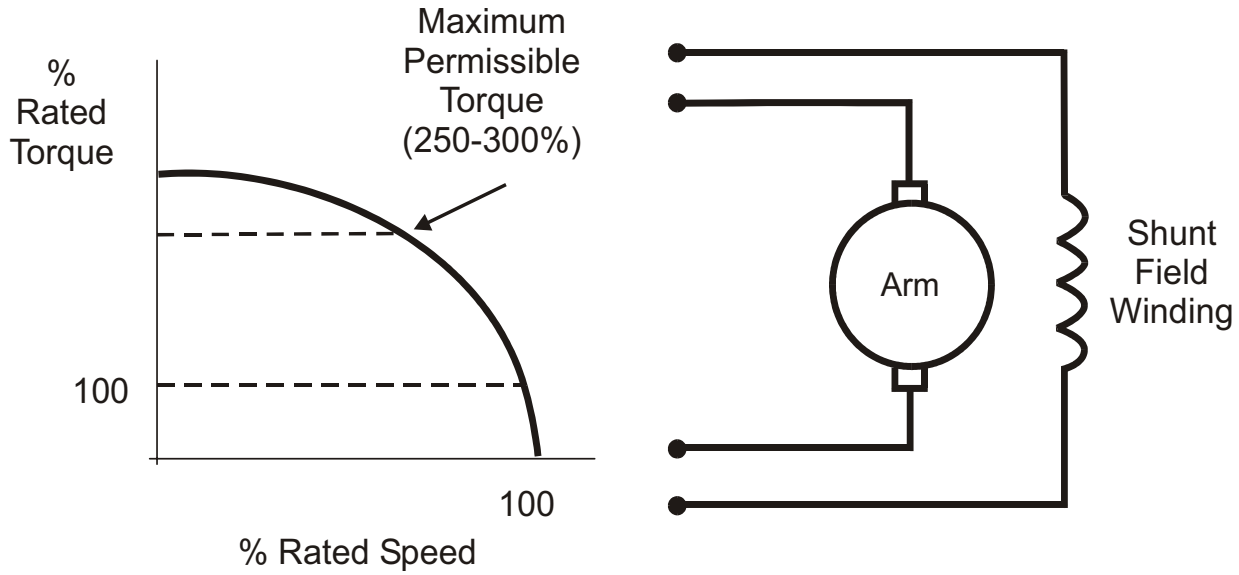


Figure 10-3: Shunt Wound DC Motor and Curve

10.3.3 Compound Wound

Compound wound DC motors are basically a combination of shunt and series wound configurations. This type of motor offers the high starting torque of a series wound motor and constant speed regulation (speed stability) under a given load. The torque and speed characteristics are the result of placing a portion of the field winding circuit, in series, with the armature circuit. When a load is applied, there is a corresponding increase in current through the series winding, which also increases the field flux, increasing torque output.

10.3.4 Permanent Magnet

Permanent magnet motors are built with a standard armature and brushes, but have permanent magnets in place of the shunt field winding. The speed characteristic is close to that of a shunt wound DC motor. This type of motor is simple to install, with only the two armature connections needed and also simple to reverse—simply reverse the connections to the armature. Though this type of motor has very good starting torque capability, the speed regulation is slightly less than that of a compound wound motor. Peak torque is limited to about 150%.

10.4 AC Motors and Their Principles of Operation

All alternating current (AC) motors can be classified into single-phase and polyphase motors (poly, meaning *many* phase or *3-phase*). For industrial applications, 3-phase induction motors are mainly used, due to efficiency. A more powerful motor can be built into a smaller frame, compared to a single-phase motor.

The main parts in an AC induction motor are the rotor (rotating element) and the stator (stationary element that generates the magnetic flux). The rotor consists of copper or aluminum bars, connected together at the ends with end rings. The rotor is filled with many individual discs of steel, called "laminations." The stator consists of cores that are also constructed with laminations. These laminations are coated with insulating varnish and then welded together to form the core (see Figure 10-4).

The revolving field set up by the stator currents cut the squirrel-cage conducting aluminum bars of the rotor. This causes voltage in these bars, with a corresponding current flow, which sets up north and south poles in the rotor. Torque (turning of the rotor) is produced due to the attraction and repulsion between these poles and the poles of the revolving stator field.

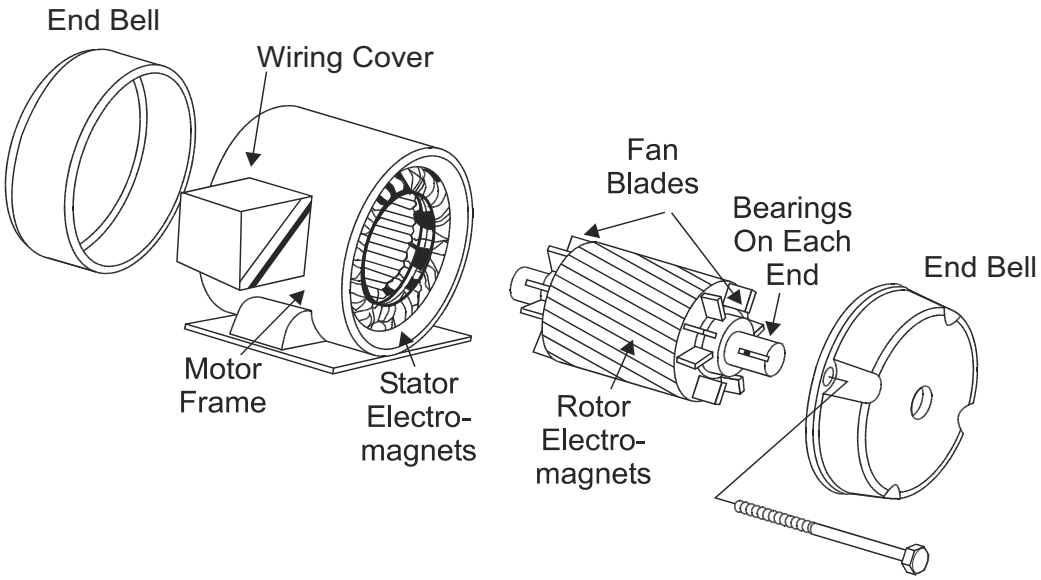


Figure 10-4: Induction Motor Construction

Each magnetic pole pair in Figure 10-5 is wound in such a way that allows the stator magnetic field to "rotate." A simple 2-pole stator shown in the figure has three coils in each pole group. (A 2-pole motor would have 2 poles x 3 phases = 6 physical poles.) Each coil in a pole group is connected to one phase of the 3-phase power source. With 3-phase power, each phase current reaches a maximum value at different time intervals. This is shown by maximum and minimum values in the lower part of Figure 10-5.

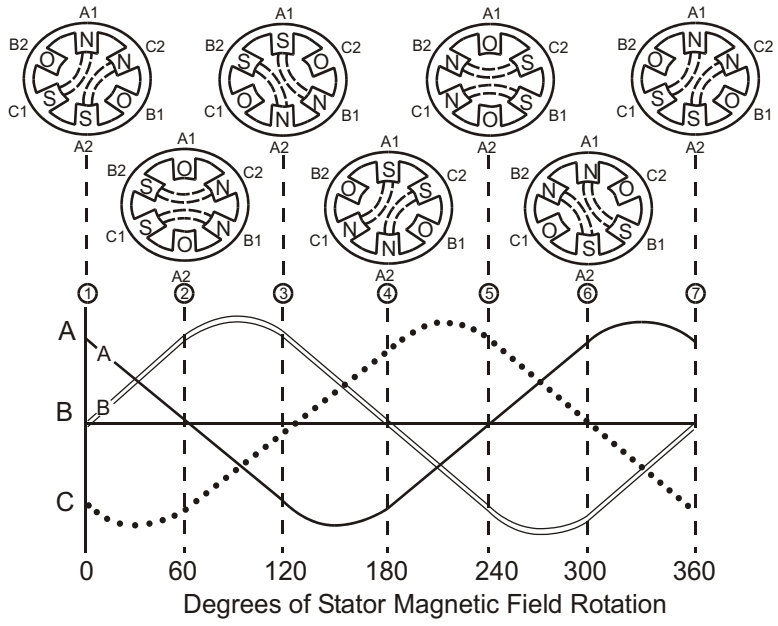


Figure 10-5: Basic Two Pole Stator

10.4.1 Control of Speed

The speed of a squirrel-cage motor depends on the frequency and number of poles for which the motor is wound (Equation 10-1).

$$N = \frac{120 \times F}{P} - \text{Slip} \tag{10-1}$$

N = Shaft Speed (RPM)

F = frequency of the power supply (Hertz)

P = number of Stator poles (pole pairs)

Squirrel-cage motors are built with the slip ranging from about 3% to 20%. The actual "slip" speed is referred to as Base Speed, which is the speed of the motor at rated voltage, rated frequency, and rated load. Motor direction is reversed by interchanging any two motor input leads.

10.4.2 Control of Torque and Horsepower (HP)

HP takes into account the "speed" at which the shaft rotates (Equation 10-2). By rearranging the equation, a corresponding value for torque can also be determined.

$$HP = \frac{T \times N}{5252} \tag{10-2}$$

T = Torque in lb-ft.

N = Speed in RPM

A higher number of poles in a motor means a larger amount of torque is developed, with a corresponding lower base speed. With a lower number of poles, the opposite would be true.

10.4.3 Enclosure Types and Cooling

The more common types of AC motor enclosures are:

- ***Open Drip-Proof Motor (ODP)*** - This enclosure type is constructed so that drops of liquid or solids falling on the machine from a vertical direction cannot enter the machine.
- ***Totally-Enclosed Non-Ventilated Motor (TENV)*** - This enclosure type is totally-enclosed and is not equipped for cooling from external devices.
- ***Totally-Enclosed Fan-Cooled Motor (TEFC)*** - This enclosure type has a shaft-mounted fan to blow cooling air across the external frame (see Figure 10-6).

10.4.4 Protection

To adequately protect the motor from prolonged overload conditions, motor overloads are installed, typically in the same enclosure as the 3-phase contactor (motor starter). These overloads (OLs) operate as "heater elements"—heating to the point of opening the circuit and mechanically disconnecting power. Overloads can be purchased with a specific time designed into the element. A Class 10 overload indicates the overload will allow 600% inrush current for 10 seconds, before opening the circuit. Class 20 would allow 600% for 20 seconds, Class 30 for 30 seconds.

An insulation system is a group of insulating materials in association with conductors and the supporting structure of a motor. Insulation systems are divided into classes according to the thermal rating of the system: Class A – (temperature up to 105° C), B – (temperature up to 130°C), F – (temperature up to 155°C), and H – (temperature up to 180°C).



Figure 10-6: Totally Enclosed Fan Cooled (TEFC) AC Motor (Courtesy of ABB Motors)

AC motors also include a voltage insulation system in the stator windings. These classes are designated by Class B, F, and H, for example. National Electrical Manufacturers Association (NEMA) standard MG1, Part 31 indicates the motor insulation voltage classes relative to use on AC drives.

10.4.5 Ratings

Because of the variety of torque requirements, NEMA has established different "designs" to cover almost every application. These designs take into consideration starting current and slip, as well as torque (see Figure 10-7).

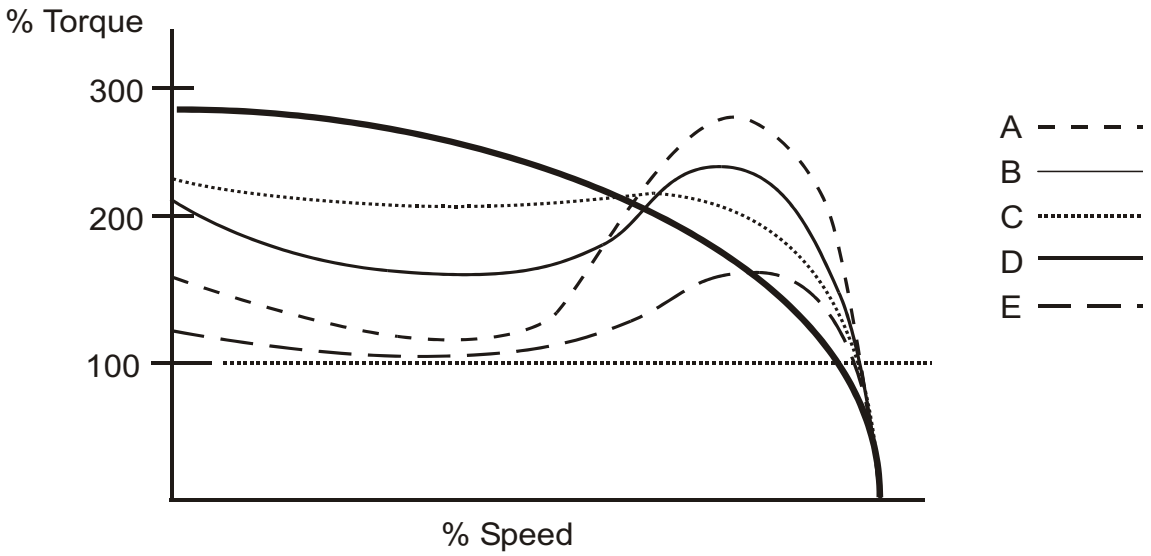


Figure 10-7: Comparison of NEMA Designs, Across the Line (Speed/Torque Characteristics)

- ***NEMA Design A*** – This type of motor has a high breakdown torque characteristic, compared to NEMA Design B motors. These motors are normally designed for specific use, with a slip characteristic usually less than 5%.
- ***NEMA Design B –*** This type of motor is designed for general purpose use, and accounts for the largest share of induction motors sold. The typical slip for a Design B motor is 3-to-5% or less.

- ***NEMA Design C*** – This type of motor has a high starting torque, with a relatively normal starting current and low slip. The type of load applied to a Design C is one where breakaway loads are high upon start.
- ***NEMA Design D*** – This type of motor has high starting torque, high slip, but also low full load speed. Because of its high slip (5-to-13%), the speed can easily fluctuate due to changes in load.
- ***NEMA Design E*** – This type of motor is known for high efficiency and is used mainly where the starting torque requirements are low (e.g., centrifugal fans and pumps).

It should be noted that in the motor world, there are two rating designations—NEMA and IEC. NEMA frame motors are in widespread use throughout U.S. industry. IEC is the acronym for the International Electrotechnical Commission. Though NEMA and IEC standards use different terms, they are essentially similar in ratings and, in many cases, are interchangeable. NEMA standards are probably more conservative, which allows for interpretations in design. IEC standards are more specific and categorized. NEMA MG-1, Part 31 standards indicate that motors operated on drives of 600V or less, should be capable of withstanding peak voltage of 1600V. Motors with a 1200V or 1000V insulation strength should not be applied to AC drives, unless additional precautions are taken.

10.5 AC Motor Types

10.5.1 Standard AC Induction

AC Motors can be divided into two major categories: asynchronous and synchronous. The induction motor is probably the most common type of asynchronous motor (meaning speed is dependent on slip). All standard motors include a small rectangular slot, cut lengthwise in the shaft, called a "keyway" or "keyseat." This slot includes a tapered cut rectangular piece of steel, called a "key," which is pressure fit into a coupler for a mechanical connection (see Figure 10-8).

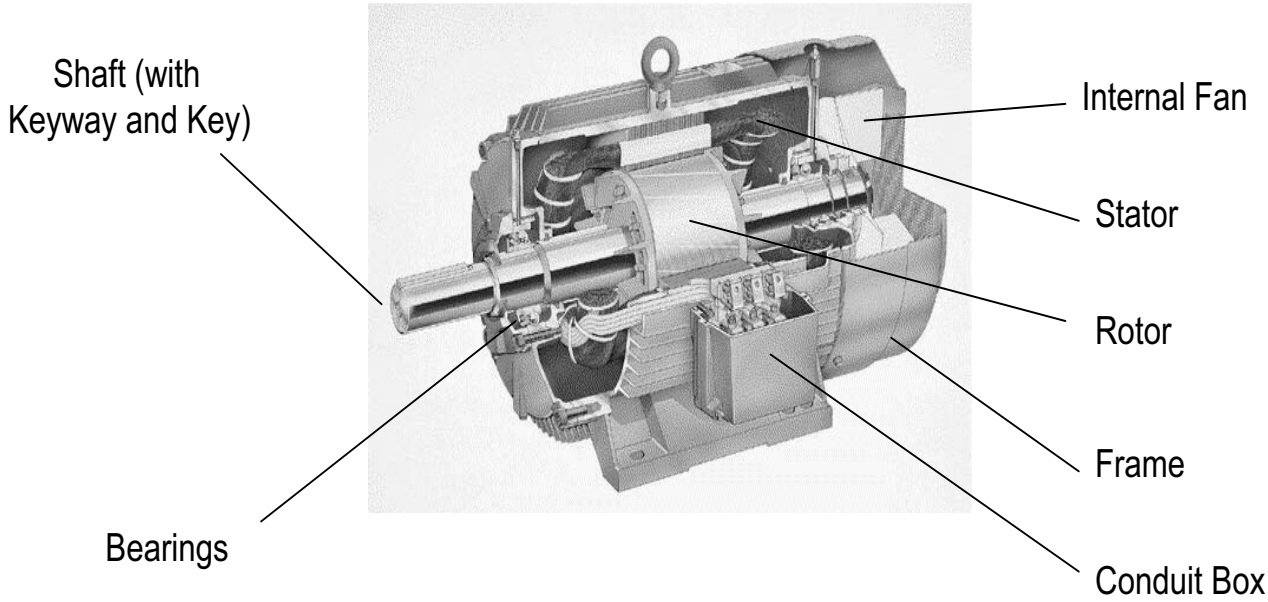


Figure 10-8: AC Induction Motor Construction (Courtesy of ABB Motors)

10.5.2 Wound Rotor

The "wound rotor" motor has controllable speed and torque characteristics. Different values of resistance are inserted into the rotor circuit to obtain various performance. They are normally started with a secondary resistance connected to the rotor circuit. The resistance is reduced to allow the motor to increase in speed. This type of motor can develop substantial torque and, at the same time, limit the amount of locked rotor current.

10.5.3 Synchronous

The two types of synchronous motors are: non-excited and DC-excited. Without complex electronic control, this motor type is inherently a fixed-speed motor. The synchronous motor could be considered a 3-phase alternator, only operated backwards. DC is applied directly to the rotor to produce a rotating electromagnetic field, which interacts with the separately powered stator windings to produce rotation. In reality, synchronous motors have little to no starting torque. An external device must be used for the initial start of the motor.

10.5.4 Multiple Pole

Multiple pole motors could be considered "multiple speed" motors. Most of the multiple pole motors are "dual speed." Essentially, the conduit box would contain two sets of wiring configurations—one for low-speed and one for high-speed windings. The windings would be engaged by electrical contacts or a two-position switch.

10.5.5 Specialty Motors (Stepper and Vector)

A "stepper" motor is one in which electrical pulses are converted into mechanical movements. A stepper motor rotates in fixed increments whenever it is "pulsed on." The size of the step, or step angle, is determined by the motor construction or by the type of controller connected. (Note: Step angle is determined in fractions of 360°). For example, the step resolution of 90° would be four steps per rev (revolution). Due to their exactness of rotation, Stepper motors are used, "open-loop," in control systems where position is critical. They are 2-phase, brushless motors that can deliver high torque at zero speed, with no drifting of the shaft position. The direction of rotation is reversed by reversing the direction of pulses from the controller.

A "vector" motor is a specific type that would be applied to an AC vector or flux vector drive. Principles of operation are identical to the standard AC induction motor. Vector control means the requirement of full torque at zero speed. The vector motor is specially designed to operate at low slip and be able to handle the heat generated by providing full torque at zero speed.

Principles of operation lie with analyzing voltage and flux vectors. The rotor is divided into 360° of rotation, which is one complete rotation. A vector would be the direction and amount of a certain quantity in the motor circuit—in this case, rotor flux or stator flux (see Figure 10-9).

Physical torque developed is a by-product of the magnitude of the stator and rotor flux vectors. Stator flux is a function of the input voltage to the motor. (The voltage vectors are indicated by U_1 to U_6 in Figure 10-9.) We could consider the dashed curve pair the torque span developed in the motor. A vector or flux vector drive would control the amount of stator and rotor flux generated. In most cases, the vector motor must have provisions for the mounting of a feedback device (encoder or resolver) on the shaft end. The feedback device sends information back to the drive control, indicating exact rotor position.

10.6 Choosing the Right Motor

10.6.1 Application Related

Both torque and speed need to be assessed for any speeds during the operation cycle. The DC motor has a higher "starting torque" capability operated from a drive. Technology advancements have

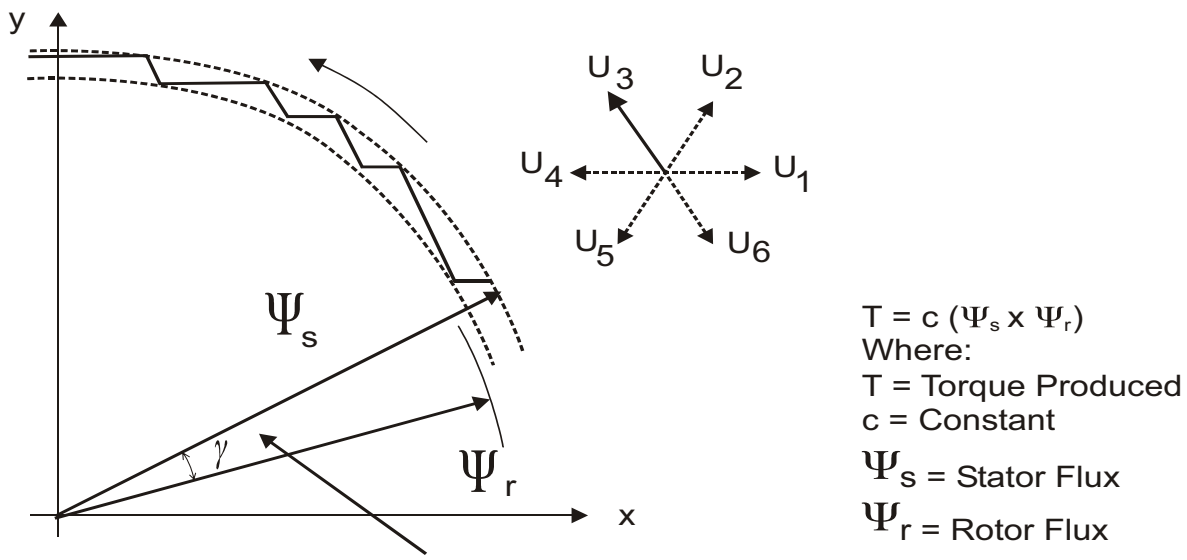


Figure 10-9: Vector Motor Relationships – Stator and Rotor Flux (Courtesy of ABB Inc.)

brought the AC motor starting torque close to that of the DC motor. Typical NEMA-D, AC motor classifications create over 300% starting torque across the line. Additional considerations relate to the environment, heat, humidity, accessibility, duty cycle, and overload requirements.

10.6.2 AC versus DC

There are no fundamental performance limitations that would prevent a flux vector variable frequency drive (VFD) from being used in any application where DC drives are used. In areas such as high speed operation, the inherent capability of AC motors exceeds the capability of DC motors. Inverter duty motors have speed range capabilities that are equal to or above the capabilities of DC motors. DC motors usually require cooling air forced through the interior of the motor in order to operate over wide speed ranges. Totally enclosed AC motors are also available with wide speed range capabilities.

Although DC motors are usually significantly more expensive than AC motors, the motor-drive package price for a VFD is often comparable to the price of a DC drive package. If spare motors are required, the package price tends to favor the VFD. Since AC motors are more reliable in a variety of situations, and have a longer average life, the DC drive alternative may require a spare motor while the AC drive may not. AC motors are available with a wide range of optional electrical and mechanical configurations and accessories. DC motors are generally less flexible and the optional features are generally more expensive.

DC motors are typically operated from a DC drive, which has reduced efficiency at lower speeds. Since DC motors tend to be less efficient than AC motors, they generally require more elaborate cooling arrangements. Most AC motors are supplied in totally enclosed housings that are cooled by blowing air over the exterior surface.

The motor is the controlling element of a DC drive system, while the electronic controller is the controlling element of an AC drive system. The purchase cost of a DC drive, in low horsepower sizes, may be less than that of its corresponding AC drive of the same horsepower. However, the cost of the DC motor may be twice that of the comparable AC motor. Technology advancements in VFD design have brought the purchase price gap closer to DC. DC motor brushes and commutators must be maintained and replaced after periods of operation. AC motors are typically less maintenance intensive, and are more “off-the-shelf” compared to comparable horsepower DC motors.

10.7 Variable Speed Drives (Electronic DC)

10.7.1 Principles of Operation

This type of drive converts fixed voltage and frequency AC to an adjustable voltage DC. A DC drive can operate a shunt wound DC motor or a permanent magnet motor. Most DC drives use silicon controlled rectifiers (SCRs) to convert AC to DC (see Figure 10-10).

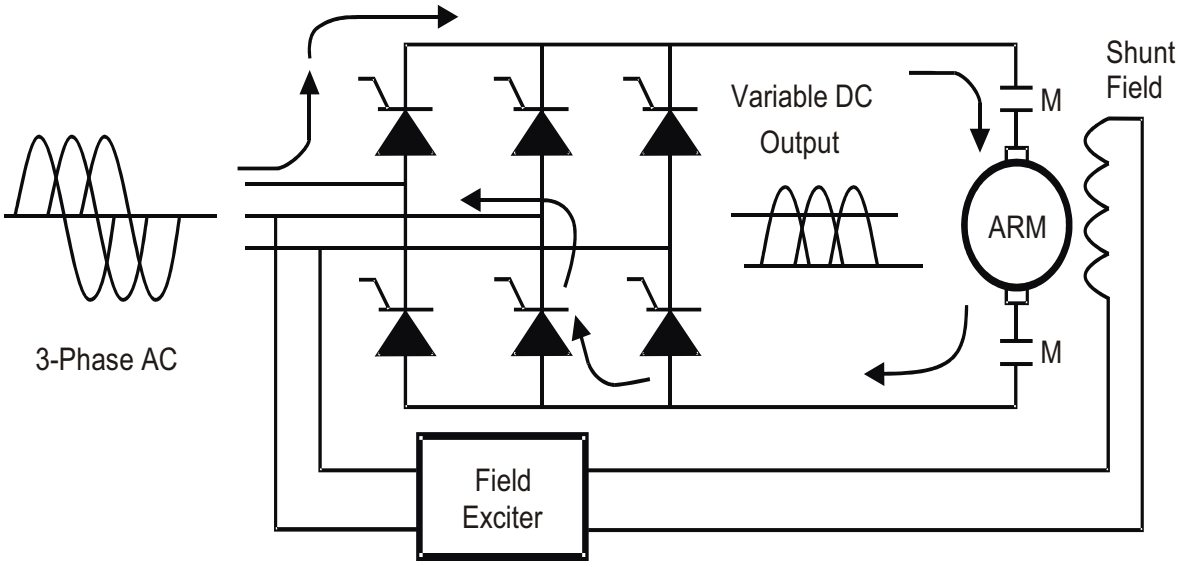


Figure 10-10: SCR Full Wave Bridge Rectification

SCRs provide output voltage when a small voltage is applied to the gate circuit. Output voltage depends on when the SCR is "gated on," causing output for the remainder of the cycle. When the SCR goes through zero, it automatically shuts off until it is gated "on" again. 3-phase DC drives, use six SCRs for full-wave bridge rectification. Insulated gate bipolar transistors (IGBTs) are now replacing SCRs in power conversion. IGBTs also use an extremely low voltage to gate "on" the device.

When the speed controller circuit calls for voltage to be produced, the M contactor (main contactor) is closed and the SCRs conduct. In one instant of time, voltage from the line enters the drive through one phase, is conducted through the SCR, and into the armature. Voltage flows through the armature and back into the SCR bridge and returns through the power line through another phase. At the time this cycle is about complete, another phase conducts through another SCR, through the armature and back into yet another phase. The cycle keeps repeating 60 times per second due to 60 Hz line input. Shunt field winding power is supplied by a DC field exciter, which supplies a constant voltage to the field winding, thereby creating a constant field flux. Many field exciters have the ability to reduce supply voltage, used in above base speed operation.

10.7.2 Control of Speed and Torque

A "speed reference" is given to the drive's input, which is then fed to the "speed controller" (see Figure 10-11).

The "speed controller" determines the output voltage for desired motor speed. The "current controller" signals the SCRs in the "Firing Unit" to "gate on." The SCRs in the converter section convert fixed 3-phase voltage to a DC voltage and current output in relation to the desired speed. The "current measuring/scaling" section monitors the output current and makes current reference corrections based on the torque requirements of the motor. If precise speed is not an issue, the DC drive and motor could

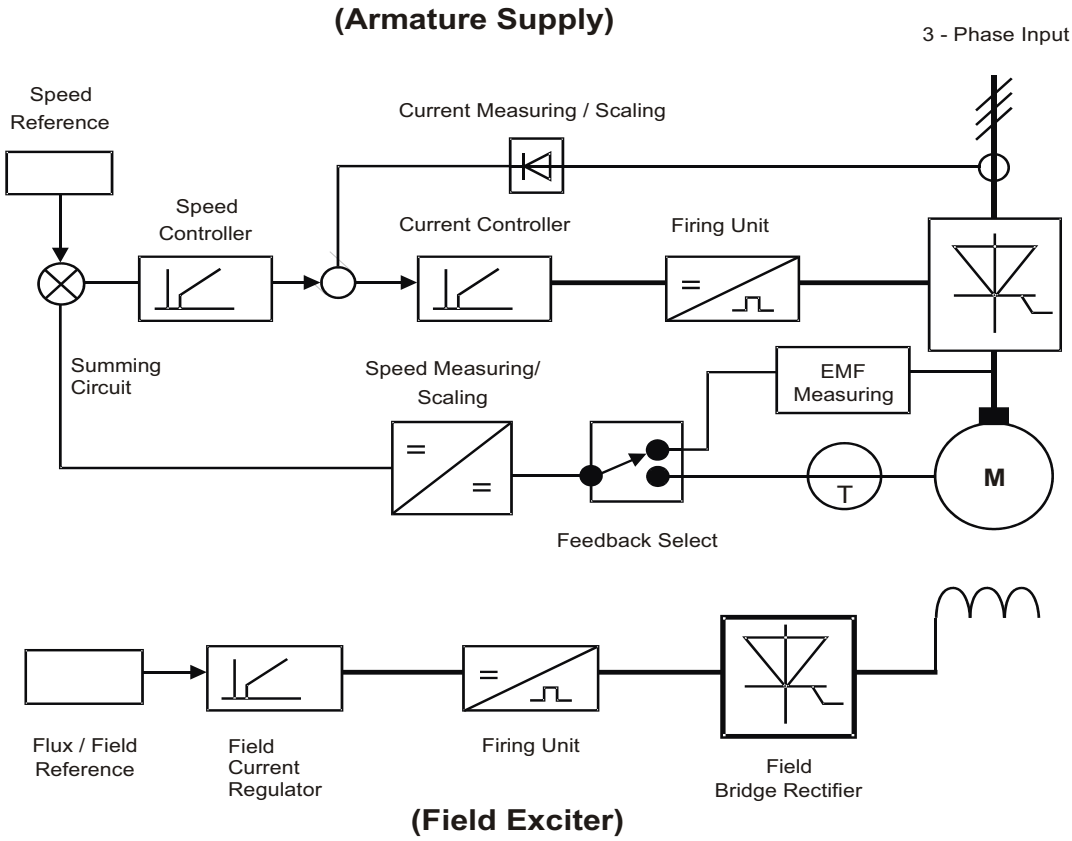


Figure 10-11: Digital Speed & Current Controllers and Field Exciter

operate "open loop." When more precise speed regulation is required, then the "speed measuring/scaling" circuit will be engaged by making the appropriate "feedback selection." If the "feedback" is using the "EMF measurement" circuit, then the "speed measuring/scaling" circuit will monitor the armature voltage output. The summing circuit will process the speed reference and feedback signal and create an "error" signal. This "error" signal is used by the "speed controller" as a new speed command—or a corrected speed command.

If tighter speed regulation is required (<1%), then "tachometer generator" feedback is required (e.g., tach feedback or tacho). A tachometer mounts on the end of the motor, opposite that of the shaft, and feeds back exact shaft speed to the speed controller. When operating in current regulation mode (controlling torque), the drive closely monitors values of the "current measuring/scaling circuit."

10.7.3 Single and 4-Quadrant Drives

There are (4) possible modes of motor operation, determined by the relationship of speed, torque and direction (Figure 10-12).

Operation in the first quadrant means the motor is actually driving the load, with torque and speed in the positive direction. This method of operation is usually accomplished with a single controller (one armature SCR bridge rectifier, and a field winding supply). The natural inertia of the system will bring the motor to a stop in acceptable time ("coast to stop").

If "coasting" to a stop is not acceptable, then a reverse polarity voltage is applied to the armature, reversing the magnetic field, and bringing the motor to a quick stop. This is called "plug" stopping. Using reverse torque to stop a motor is called "two quadrant operation."

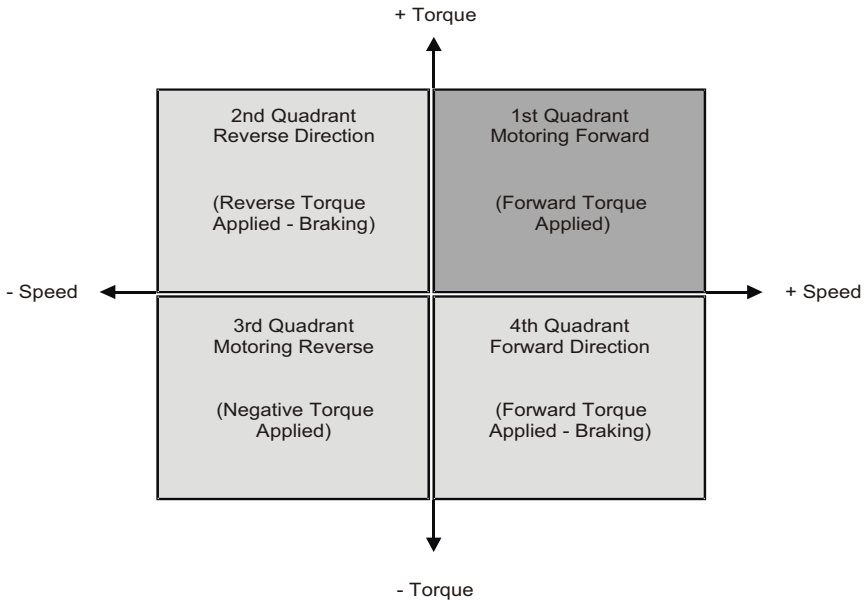


Figure 10-12: Single, Two and Four-Quadrant Operation

If true control throughout the positive and negative speed and torque range is desired, then a 12 SCR bridge controller is required (two armature SCR bridge rectifiers—one in the forward and one in the reverse direction), as is a field winding supply. The motor can be brought to a very fast stop by engaging the reverse armature supply bridge and "regenerating" the motor's energy back into the power line. This is called "regenerative braking." A 4-quadrant supply (4-quadrant operation) is used if "motoring" the load in the reverse speed and torque direction is required. The "reverse" SCR bridge is now used as the driving supply and the "forward" SCR bridge acts as the braking device to bring the motor quickly back to zero speed.

10.7.4 Braking Methods (Dynamic and Regenerative)

After "ramp to stop," the next fastest stopping time would be achieved by dynamic braking (see Figure 10-13).

This form of stopping uses a fixed, high wattage resistor (or bank of resistors) to transform the rotating energy into heat. This system uses a contactor to connect the resistor across the DC bus at the time needed to dissipate the voltage generated by the motor.

The fastest "electronic" stopping method is that of "regeneration." With "regenerative braking," all of the motor's energy is fed directly back into the AC power line. In order to accomplish this, a second set of "reverse connected" SCRs is required. This allows the drive to conduct current in the opposite direction (generating the motor's energy back to the line). A regenerative drive allows "motoring" and "regenerating" in both the forward and reverse directions.

10.8 Variable Speed Drives (Electronic AC)

10.8.1 Principles of Operation – Pulse Width Modulation (PWM)

There are several types of AC drives (VFDs—Variable Frequency Drives). All of them have one concept in common: they convert fixed voltage and frequency input into a variable voltage and frequency output to change the speed of a motor (see Figure 10-14).

3-phase power is applied to the input section of the drive, called the "converter." This section contains (6) diodes, arranged in an electrical "bridge." These diodes convert AC power to DC power. The "DC

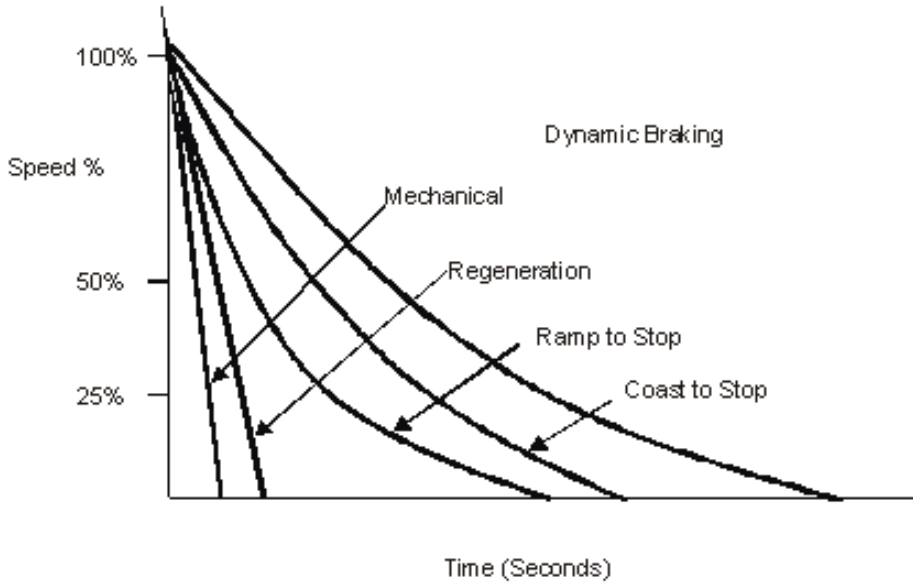


Figure 10-13: Braking Methods for DC Drives

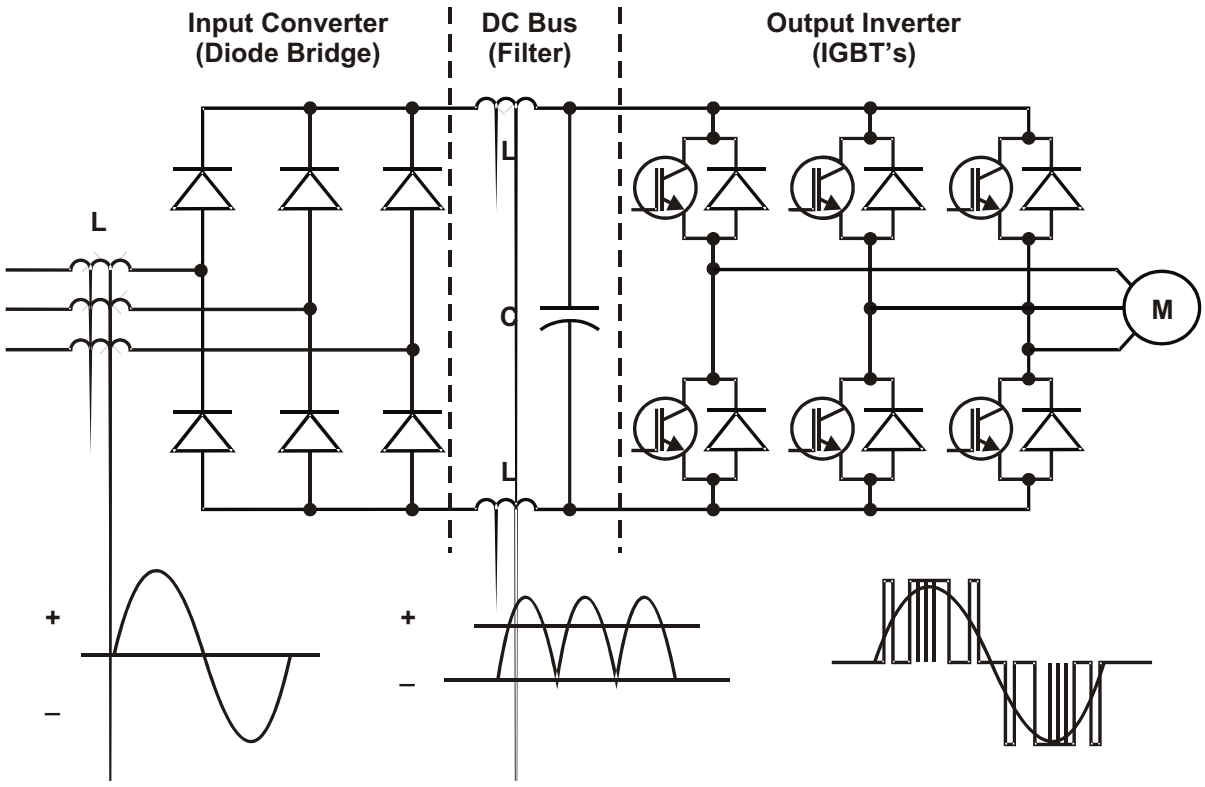


Figure 10-14: PWM Drive (VFD) Block Diagram (Courtesy of ABB Inc.)

bus" section accepts the now converted, AC to fixed voltage DC. The "DC bus" filters and smoothes the waveform, using "L" (inductors) and "C" (capacitors). The diodes reconstruct the negative halves of the waveform onto the positive half, with an average DC voltage of approximately 650-680 Volts (460VAC unit).

Once filtered, the DC bus voltage is delivered to the final section of the drive, called the "inverter" section. This section "inverts" DC voltage back to AC—but in a variable voltage and frequency output. Devices called insulated gate bipolar transistors (IGBTs) act as power switches that turn on and off the DC bus voltage, at specific intervals. Control circuits, called gate drivers, cause the control part of the IGBT (gate) to turn "on" and "off" as needed.

10.8.2 Control of Speed and Torque

The torque of a motor is determined by a basic characteristic—the volts per Hertz ratio (V/Hz). If an induction motor is connected to a 460 volt power source, at 60 Hz, the ratio is 7.67 V/Hz. As long as this ratio is kept in proportion, the motor will develop rated torque. The output of the drive doesn't actually provide an exact replica of the AC input sine waveform (see Figure 10-15).

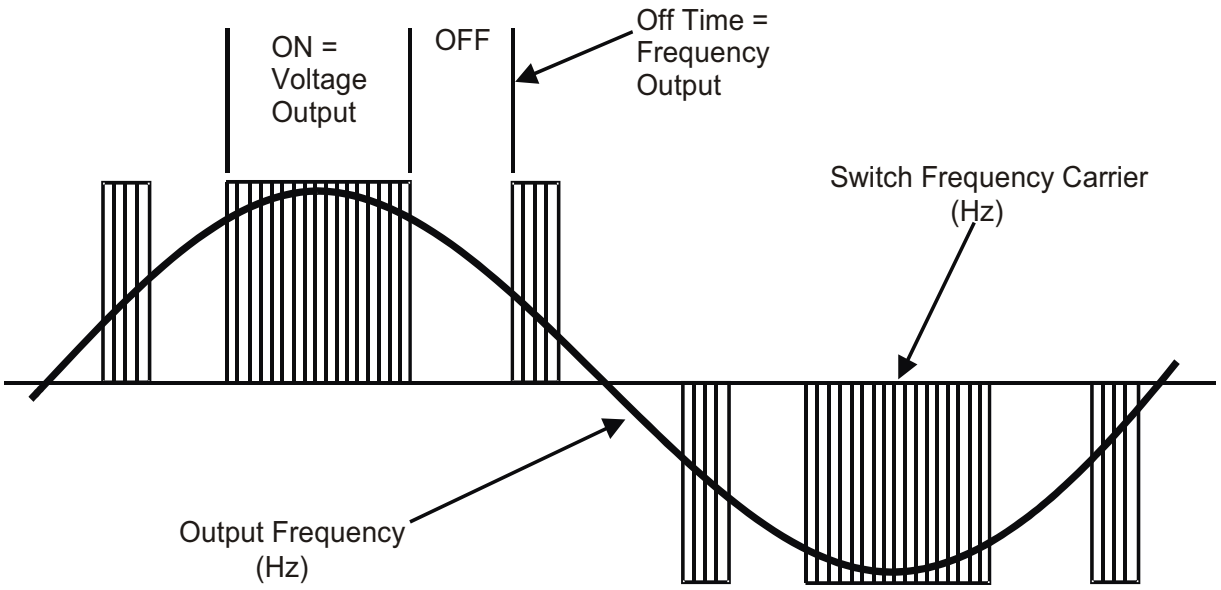


Figure 10-15: Frequency and Voltage Creation from PWM

It actually provides voltage pulses that are at a constant magnitude in height. The positive and negative switching of the IGBTs re-creates the 3-phase output. The speed at which IGBTs are switched is called the "carrier frequency" or "switch frequency." The higher the switch frequency, the more resolution each PWM pulse contains (typical carrier frequencies range from 3kHz to 16kHz).

10.8.3 Other VFD Types (Variable Voltage Inverter, Current Source Inverter, Flux Vector, Sensorless Vector, Torque Controlled)

A VVI design takes the supply voltage, rectifies it using controllable SCRs and sends the variable voltage to the DC bus and then to the inverter section. The inverter section then "inverts" the variable voltage DC to a variable voltage and frequency AC. The inverter section contains power semiconductors such as transistors or thyristors (SCRs).

A CSI drive has components similar to the VVI drive. The major difference is that it is more of a current-sensitive drive, as opposed to a VVI which is more voltage-sensitive. This design also takes the supply voltage, rectifies it, and sends the variable voltage to the DC bus and then to the inverter. The inverter section "inverts" variable voltage DC to a variable voltage and frequency AC. The inverter section is made up of power semiconductors such as transistors or thyristors (SCRs).

One of the basic principles of a flux vector drive is to simulate the torque produced by a DC motor (full torque at zero speed). Until the advent of flux vector drives, slip had to occur [30 to 50 revolutions per minute (RPM)], in order for motor torque to be developed (termed a V/Hz or scalar drive). With flux vector control, the drive forces the motor to generate torque, at zero speed.

A flux vector drive features field-oriented control similar to that of a DC drive, where the shunt field windings continuously have flux, even at zero speed. The motor's electrical characteristics are simulated in the drive controller circuitry called a "motor model." The motor model takes a mental impression of the motor's flux, voltage, and current requirements for every degree of shaft rotation. To emulate the magnetic operating conditions of a DC motor, the flux-vector drive needs to know the angular position of the rotor flux. The rotor status is fed back to the drive logic by a "pulse encoder" and a microprocessor is used to mathematically model and process the data. The advantages of this type of drive include: good torque response (<10 msec) and full torque at zero speed (at approximately 0.5 Hz output).

Sensorless flux vector control is similar to a DC drive's electromagnetic field (EMF) control. With sensorless flux vector control, a modulator is used to vary the strength of the field, which is in reality, the stator.

The role of sensorless flux vector fits generally in between the standard PWM open loop control method and a full flux vector, closed loop control method. This method provides higher starting and running torque, as well as smoother shaft rotation at low speed, compared with standard V/Hz, PWM drives. One of the main advantages of sensorless flux vector over standard PWM is higher starting torque on demand.

The direct torque control method is similar to an AC sensorless vector drive, which uses a *direct torque control* scheme. Field orientation is achieved without feedback using advanced motor theory to calculate the motor torque directly. There is no modulator used in direct torque control and no need for a tachometer or position encoder for speed or position feedback of the motor shaft. This drive has a torque response that is as much as 10 times faster than any AC or DC drive. The dynamic speed accuracy of this drive is many times better than any open loop AC drive. It is also comparable with a DC drive that uses feedback.

Direct torque control includes the basic building blocks upon which the drive does its calculations, based on a motor model (see Figure 10-16).

The two fundamental sections of direct torque control are the *torque control loop* and the *speed control loop*. During drive operation, two output-phase current values and the DC bus voltage value are monitored, along with the IGBT switch positions. This information is fed to the *adaptive motor model*. The motor model calculates the motor data on the basis of information it receives during a self-tuning process (motor identification). During this automatic tuning process, the drive's motor model gathers information, such as stator resistance, mutual inductance, and saturation coefficients, as well as the motor inertia.

The output of this motor model is the representation of actual motor torque and stator flux for every calculation of shaft speed. The values of actual torque and flux are fed to their respective comparators, where comparisons are performed every 25 μs. Every 25 μs, the inverter IGBTs are sent optimum pulse information for obtaining accurate motor torque. The correct IGBT switch combination is determined during every control cycle.

10.8.4 Braking Methods (Dynamic and Regenerative)

The DC bus of a typical AC drive will take on as much voltage as possible, without tripping. If an overvoltage trip occurs, the operator has three choices—increase the deceleration time, add "DC injection braking" (a parameter), or add an external dynamic braking package. If the deceleration time is extended, the DC bus has more time to dissipate the energy and stay below the trip point.

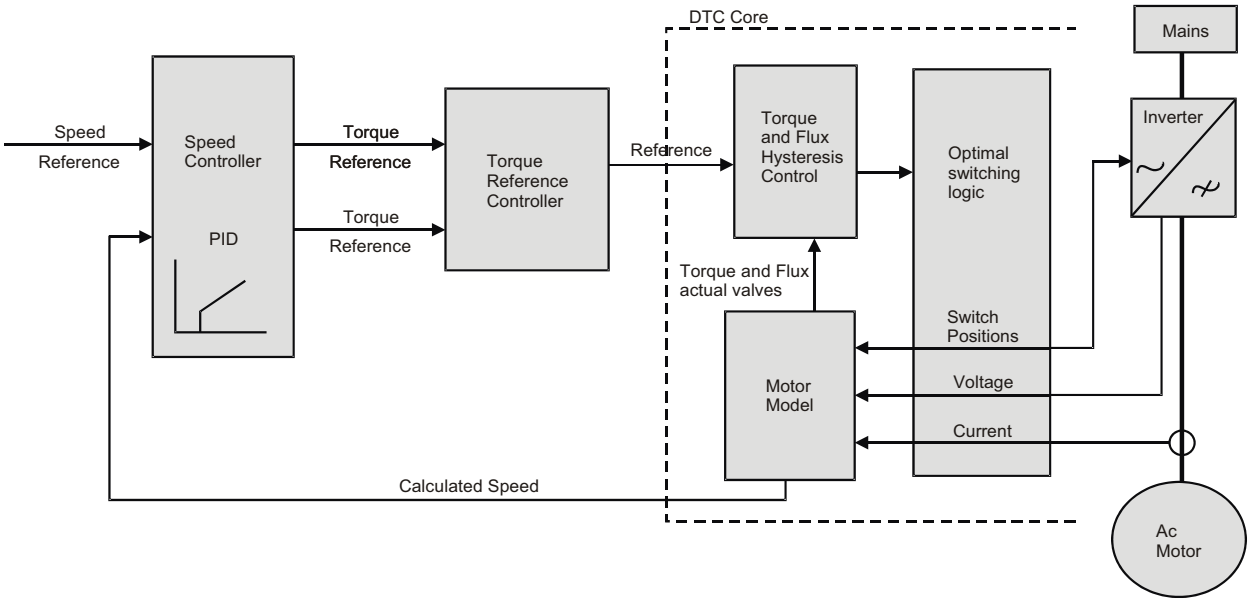


Figure 10-16: The Direct Torque Control (DTC™) Method (Courtesy of ABB, Inc.)

With DC injection braking, DC voltage is "injected" into the stator windings for a pre-set period of time. Braking torque (counter torque) brings the motor to a quicker stop, compared to "ramp." Dynamic braking (DB) uses an externally mounted fixed, high wattage resistor (or bank of resistors) to transform the rotating energy into heat (see Figure 10-17).

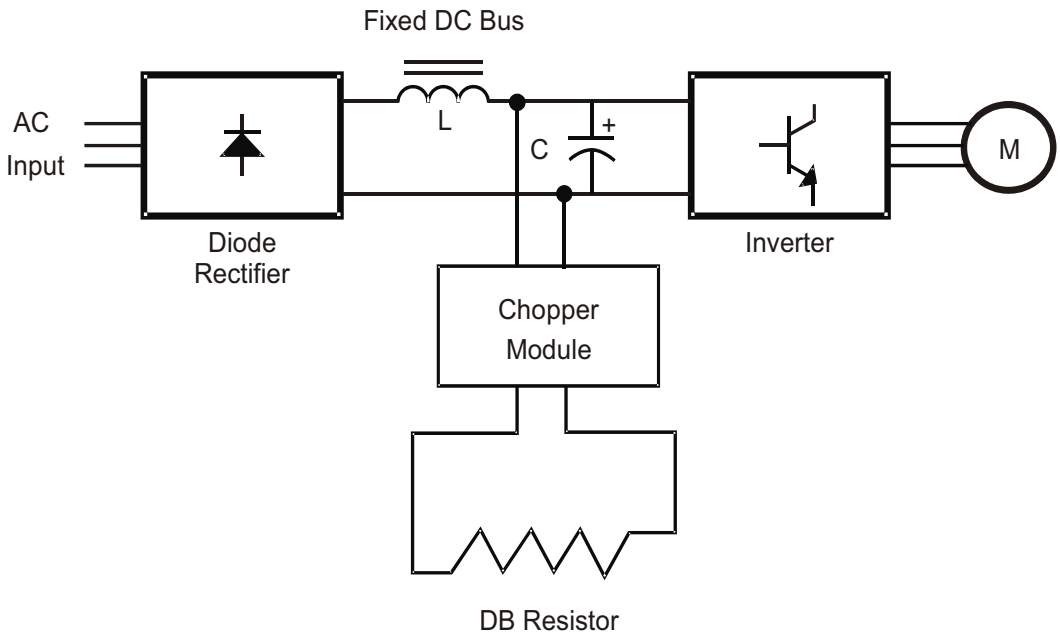


Figure 10-17: AC Drive Dynamic Braking

When the motor is going faster than commanded speed, the rotational energy is fed back to the DC bus. Once the bus level increases to a pre-determined point, the "chopper" module activates, and the excess voltage is transferred to the DB resistor.

For regenerative braking, a second set of "reverse connected" power semiconductors is required. The latest AC regenerative drives use two sets of IGBTs in the converter section (some manufacturers term this an "active front end"). The reverse set of power components allows the drive to conduct current in the opposite direction (taking the motor's energy and generating it back to the line). As expected with a 4-quadrant system, this unit allows driving the motor and regenerating in both the forward and reverse directions.

10.9. Automation and the Use of VFDs

The more complex AC drive applications are now accomplished with modifications in drive software, which some manufacturers call "firmware." IGBT technology and high-speed application chips and processors have made the AC drive a true competitor to that of the traditional DC drive system.

10.9.1 Intelligent and Compact Packaged Designs

Because of the use of microprocessors and IGBTs, a 1 HP drive of today is about one-third the size of a 1 HP drive 10 years ago. This size reduction is also attributed to "surface mount" technology used to assemble components to circuit boards. AC drive designs have fewer parts to replace, and include troubleshooting features available through "on-board" diagnostic or "maintenance assistant" software. In most cases, packaged AC drives of approximately 50 HP or less only use two circuit boards—control board and motor control board.

Programming is typically done with a removable touch keypad or remote operator panel. With the latest advancements in E^2PROMs and "flash PROMs," the programming panel can be removed from power after permanently storing parameter values. Drive panels guide the user through use of a "start-up assistant" and feature multi-language programming and "soft keys" similar to that of a cell phone. The functions of the keys change depending on the mode of the keypad. Keypads also feature pre-programmed default application values called "macros" such as PID (proportional-integral-derivative), as shown in Figure 10-18.

10.9.2 Serial and Fiber Optic Communications

Control and diagnostic data can be transferred to the upper level control system at a rate of 100 milliseconds. With only three wires for control connections, the drive "health" and operating statistics are available through any connected laptop.

Fiber optic communications use plastic or silica (glass fiber) and an intense light source to transmit data. With optical fiber, thousands of bits of information can be transmitted at a rate of 4 million bits per second (4M Baud). Several drive manufacturers offer serial and fiber optic software that installs directly onto a laptop or desktop computer, giving access to all drive parameters.

10.9.3 Fieldbus Communications (PLCs)

Data links to programmable logic controllers (PLCs) are common in many high-speed systems that process control and feedback information. Several manufacturers of PLCs offer a direct connection to many drive products. Because each PLC uses a specific programming language, drive manufacturers are required to build an "adapter" box (called a fieldbus module) to translate one language to another (called a protocol). Several manufacturers allow drive connections to existing internal network structures through the use of Ethernet modules. Modules that communicate through TCP and IP addresses allow high level controls through automated internal systems. Additional interfaces through wireless technologies and personal digital assistants (PDAs) are also making their way into programmable VFDs.

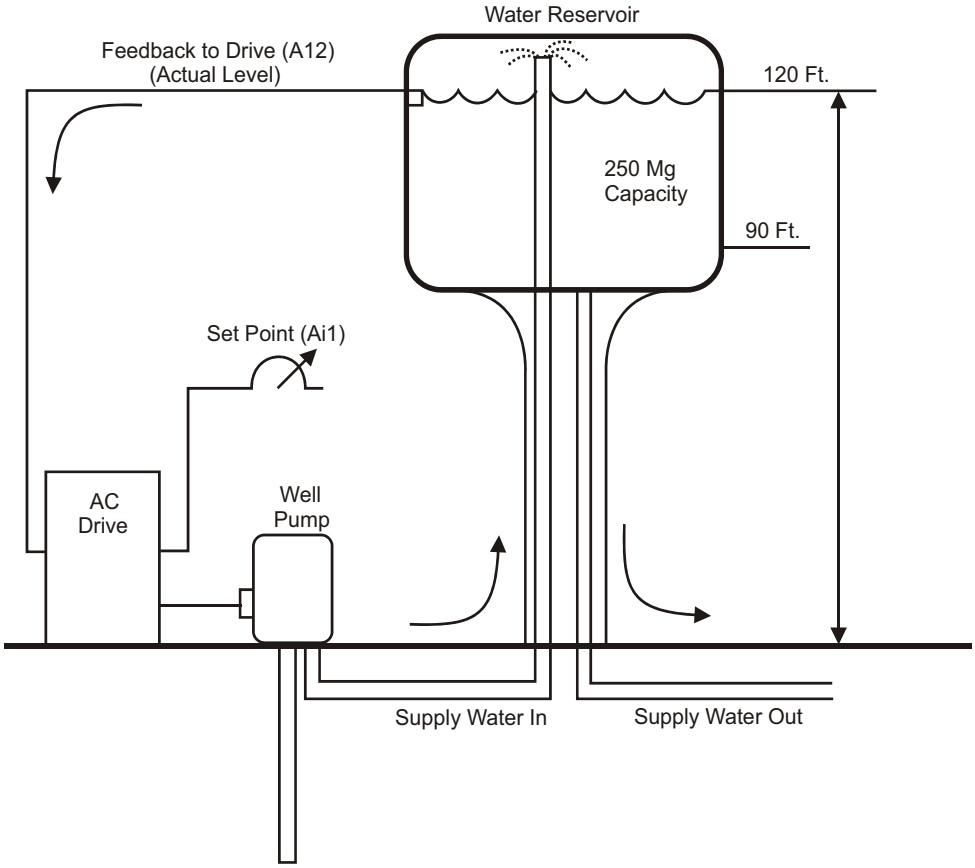


Figure 10-18: PID Control in a Pumping Application

10.9.4 Drive Configurations

Several manufacturers offer a variation of the standard 6-pulse drive. An AC drive that is termed "12-pulse ready" offers the optional feature of converting a standard 6-pulse drive to a 12-pulse drive (addition controls and a phase-shift transformer are required). The 12-pulse drive does an impressive job of reducing the highest contributors of harmonic distortion back to the power line.

One of the features of AC drive technology is the ability to "bypass" the drive if it stops operating for any reason. Known as "bypass," this configuration is used in many applications where a fan or pump must continue operating, even though it is at fixed speed. One manufacturer, ABB Inc., offers "E-bypass" circuitry. If required, a circuit board operates all the diagnostics and logic for an "automatic" transfer to bypass and feeds bypass information to the building automation system.

10.9.5 Chapter Summary

The use of electronic motor speed controls is expected to grow in non-traditional applications, such as automotive subsystems, household appliances, electric vehicles, people movers and marine propulsion units. As of this printing, VFDs comprise over 50% of the control methods used on standard induction AC motors. That trend will only increase in the years to come, due to increased focus on energy saving devices and micro-drive use (below 5 HP). Trends in AC drives appear to include: drives with increased intelligence; extensive use of communication options; and the use of external control devices for drive information, I/O and feedback status. Future technology will allow for an identification (ID) chip to be embedded into the motor, allowing for an almost "automated" start-up.

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